Climatic Change (2025) 178:88
https://doi.org/10.1007/s10584-025-03930-7

From risk to resilience: geospatial decision support tool and adaptation strategies for coastal flooding in Rodrigues Island

Jay Rovisham Singh Doorga[1] · Olivier Pasnin[2] · Yeshna Dindoyal[3]

Received: 19 May 2024 / Accepted: 9 April 2025 / Published online: 23 April 2025
© The Author(s), under exclusive licence to Springer Nature B.V. 2025

Abstract
Rodrigues Island, being a small Island Developing State (SIDS) in the Indian Ocean, is increasingly vulnerable to the impacts of climate change, particularly rising sea levels and recurrent storm surges. The motivation for this study stems from the need to safeguard Rodrigues' socio-economic activities, which are intricately linked to its geophysical landscape and ecosystem services. This study employs a geospatial-based approach to identify the coastal areas most at risk and to provide a foundation for tailored adaptation strategies. Using predefined scenarios established by the Intergovernmental Panel on Climate Change (IPCC) and storm surge models, the research projects sea-level rise for Rodrigues over the next century. Approximately 1.45 km^2 of terrestrial land are exposed to flooding under SSP5-8.5, which represents a loss equivalent to above 1.34% of the island's total land territory imperiled by the encroaching and rising sea levels. This will induce infrastructural damage reaching 23.6%. The results additionally reveal that the low-lying coastal regions at risk are Rivière Cocos, Baie Aux Huitres, Anse Mourouk, Grand Baie, and Port Mathurin. The research also highlights that as these vulnerable areas are progressively inundated, increased human migration may become an unavoidable adaptation strategy for the affected communities. The study emphasizes the importance of innovative and multifaceted solutions, including nature-based strategies such as mangrove reforestation, coral reef restoration, and terrestrial conservation, alongside the establishment of Marine Protected Areas and the implementation of advanced coastal engineering techniques. Additionally, the research advocates for a thorough review and enhancement of the legal framework, with particular focus on relocation strategies. This study contributes to a better understanding of climate change impacts on Rodrigues Island and provides a critical basis for policy development.

Keywords Coastal flooding · Sea level rise · Climate change · SIDS · Rodrigues · Indian Ocean

Extended author information available on the last page of the article

1 Introduction

The Intergovernmental Panel on Climate Change (IPCC) (Fox-Kemper et al. 2021) reports a global Sea-Level Rise (SLR) of 0.31 ±0.07 cm/year, attributing 90% of this increase to climate change, leading to thermal expansion, and glacier and ice sheet melting. The rising global mean sea level (GMSL) and associated changes in sea-level extremes present considerable risks to low-lying coastal communities worldwide (McGranahan et al. 2007; Magnan et al. 2022; Fox-Kemper et al. 2021). These extreme sea levels result in various coastal hazards, including land submergence, increased flooding, coastal erosion, ecosystem alterations, soil and water salinization, hindered drainage, and economic losses, which particularly threaten small islands (Fox-Kemper et al. 2021). SLR also impacts both terrestrial and marine ecosystems, affecting biodiversity and human populations, especially in developing regions, and causing considerable population displacements (McGranahan et al. 2007; Lincke and Hinkel 2021). Historically, storm surges from tropical cyclones have caused damage and loss of life in low-lying coastal areas (Needham et al. 2015). Over the past four decades, tropical cyclones have resulted in over USD 2100 billion in damages, with 90% of human and property losses due to cyclonic floods (Pielke et al. 2008). Knutson et al. (2020) predict a 13% increase in Category 4 and 5 tropical cyclones with a 2 °C rise in global temperature.

The vulnerability of low-lying coastal areas is exacerbated by the combined impacts of SLR, increased precipitation, and intensified storms, which collectively raise the risk of flooding for coastal communities (Fox-Kemper et al. 2021). This situation challenges SIDS, which are vulnerable geographical entities (Doorga et al. 2024, 2021; Mycoo 2018). They are often described as being on the frontlines of climate change, hot spots of climate change, or canaries in the coalmine (Farbotko 2010; Walshe and Stancioff 2018; Thomas et al. 2020). Defined by their limited landmass, low-lying topography, and susceptibility to climate change impacts and burdened disproportionately by climate change (Lee and Romero 2023), SIDS are particularly vulnerable to the adverse effects of rising sea levels, extreme weather events, and changing ocean dynamics (Fox-Kemper et al. 2021; Magnan et al. 2022). While it is not denied, that there has been a growing academic focus on understanding the impacts and ramifications of climate change within SIDS (Scandurra et al. 2018) recent studies by Klöck and Nunn (2019) and Petzold and Magnan (2019) underscore a gap in the existing literature: the absence of a comprehensive and systematic review concerning climate change adaptation tailored to SIDS, along with a lack of literature addressing the evolving conceptions of vulnerability, resilience, and transformation in these regions, both pre- and post- 5th Assessment Report (AR5) (Robinson 2020). Klöck and Nunn's (2019) approach which delineated the principal climate impacts necessitating adaptation in SIDS, encompassed literature published between 2000 and 2016.

However, despite these contributions, the plea articulated by Wong (2011) for an urgent need for a comprehensive assessment consisting of existing knowledge, experiences, data gaps and future needs specific to the SIDS group remains unaddressed. Additionally, in the absence of adequate adaptation measures, the majority of low-lying islands, coastal regions, and communities confront considerable risks stemming from these coastal hazards (Fox-Kemper et al. 2021), and while adaptation in SIDS necessitates urgent attention, it remains complicated (Saxena et al. 2018).

This study seeks to investigate the impacts of climate change on coastal low-lying areas exposed to flooding, for a small Indian Ocean Island, Rodrigues, which forms part of the Republic of Mauritius. The combined impact of sea-level rise, storm surge, and peak tides are investigated based on local climate and environmental conditions using predefined scenarios established by the IPCC (Fox-Kemper et al. 2021). This is achieved by aggregating the predicted storm surge levels, sea-level rise, and astronomical tidal conditions over the next century from 2030 to 2130. Surge inundation models serve as important tools for exploring the potential ramifications of various flooding scenarios, particularly those representing worst-case outcomes, and are instrumental in informing the development of effective adaptation strategies that will fill critical gaps. Drawing upon Tadesse et al. (2020) and Doorga et al. (2024), this study endeavors to extend the understanding of the effects of sea-level rise and changes in storm intensity in a specific locale situated within the Indian Ocean. Existing research on vulnerability and risk assessment of coastal areas exposed to sea-level rise and flooding remains limited in Rodrigues, while studies on holistic coastal flooding risk assessments at local scales, is scarce (UNDP 2021).

To address these, the main purpose of this study is to assess flood risk due to SLR and coastal flooding by deploying a GIS-based coastal flood risk analysis and to identify highly vulnerable coastal areas around Rodrigues Island. This study provides, for the first time, storm tide projections from 2030 to 2130 for five Shared Socio-economic Pathways (SSPs) scenarios for Rodrigues Island. It also identifies flood prone areas along the coast as a result of SLR and coastal flooding forcings such as storm surge and tides. Lastly, this work provides guidance for policymakers to help the formulation of adaptation policies in Rodrigues and other SIDS. Wang et al. (2021) undertook a comprehensive risk assessment of storm surge along a chosen coastal site with different scenarios derived from five different typhoon intensities. Furthermore, they conducted a vulnerability assessment based on the land cover types present along the coastal region selected. Overall, they combined hazard assessment with vulnerability assessment to produce a mapping for potential storm surge risk. In another study by Glas et at. (2017), a coastal town was chosen as study site where a risk-based tool was developed using parameters such as flood data, land use, socioeconomic and rainfall values to estimate damage from fresh and saltwater. A study done in Italy (Apollonio et al. 2020) assessed the economic risk of a coastal area using four different flooding scenarios through storm surge.

Most SIDS have initiated a formal stance towards adaptation through their Nationally Determined Contribution (NDC) and National Adaptation Plan (NAP) (McMichael and Katonivualiku 2020). This paper seeks to contribute to the updated NDC of the Republic of Mauritius which dedicates 4.5 USD billion for adaptation and which includes a new policy orientation in key adaptation sectors including Coastal Zones (Mauritius NDC 2021). By centering attention on a delimited region, like Rodrigues, within a particular oceanic basin, the issue of coastal flooding can be meticulously examined and this methodological choice and localized approach ensures not only the contextual relevance of findings but also their practical applicability. Insights are crucial for understanding the unique regional challenges and opportunities, providing a foundation for policy development, decision-making, and tailored adaptation strategies. These insights help address specific needs and nuances of the region, serving as a model for other SIDS to draw inspiration from and adapt according to their vulnerabilities.

In order to bridge this considerable knowledge, literature, and data gap, and address the above-mentioned concerns, the current study therefore seeks to, *inter alia*: (1) identify low-lying areas exposed to flooding in Rodrigues; (2) propose long-term adaptation measures and placed-based management strategies for coastal resilience.

2 Study area

Rodrigues Island (19° 43′ S, 63° 25′ E) is located at about 1,440 km east of Madagascar and is the smallest of the Mascarene archipelago situated in the Indian Ocean (Fig. 1). It is inhabited by approximately 43,650 people. The island is primarily of volcanic origin and measures approximately 18.3 km long and 6.5 km wide and reaches a maximum height of 396 m at Mt Limon. The island stretches over a surface area of 108 km^2. The coral reef around the island measures about 70 km in length, enclosing a total lagoon area of 220 km^2 about twice its land extent, and serving as an effective natural barrier from waves. Rodrigues is surrounded by a fringing reef which is nearly continuous around the island covering a total area of approximately 230.6 km^2 (Turner and Klaus 2005). The lagoon flushes completely in four to seven days (Turner et al. 2000) aided by three major tunnels and several small passes in the reef (Winton 2006). The tidal range within the lagoon is a maximum of 1.5 m (Hardman et al. 2006). GIS mapping of biotopes revealed that the lagoon is dominated by sand (40%), vegetation which is mainly seagrass and macro-algae (28%), and lagoon corals (21%) (Chapman and Turner 2004).

The Rodriguan economy is based on subsistence agriculture, cattle rearing and fishing. However, since the 1980's, Rodrigues moved from a subsistence economy towards a diversified economy by developing new sustainable sectors such as tourism, information and communication and small-scale industry, among others (Agathe 2024). The Rodrigues Regional Assembly is investing into the extension of the runway to accommodate larger aircraft types on the island thus improving the tourism sector in Rodrigues. With the rate of sea-level rise in Rodrigues estimated at 9.0 mm/year, higher than that of Mauritius at 5.0 mm/year (Ragoonaden et al. 2017); careful consideration should be taken for a sustainable development and long term development of the tourism sector while also considering the rise in sea-level.

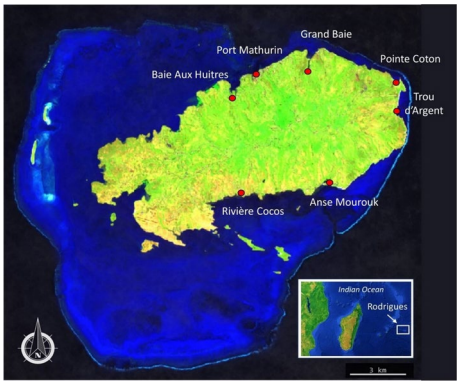

Fig. 1 Rodrigues Island, located in the South West Indian Ocean (SWIO) basin

3 Methods

3.1 100-year storm tide

The method used for analyzing storm tide level in Rodrigues is based on the cumulative contributive assessments of predicted storm surge levels, sea-level rise, and astronomical tidal conditions over the next century from 2030 to 2130. Losses and damages incurred in Southern Indian Ocean Islands due to high storm surge events are mostly attributed to cyclonic weather conditions (Doorga et al. 2024). The return periods of the storm surges in the vicinity of Rodrigues Island are estimated based on data sourced from the Global Storm Surge Reconstruction (GSSR) model. The surge model developed by Tadesse et al. (2020) reconstructs daily maximum surge values at respective tide gauge locations using multiple linear regression and Random Forest. The reconstructed value at a gauge is given as follows:

$$Surge\,(t,d) = a + \sum\nolimits_{i=1}^{N} \sum\nolimits_{j=1}^{M} b_{ij} \times PC_{ij}\,(t,d) \tag{1}$$

where $Surge\,(t,d)$ is the maximum daily surge at the t-th gauge on specific day, d; PC_{ij} denotes the *j*-th principal Principal Component (PC) of the *i*-th predictor of that day; *M* and *N* represents the total number of predictors. The regression coefficients of the fitted model are given by a and b_{ij}. Training and validation of data for model reconstruction were conducted using Random Forest.

Best fit linear and polynomial regression models are thereafter generated (Fig. 2a) using a Spearman Rank Correlation Coefficient (SRCC) method which maps the return period (logarithmic horizontal scale) to the surge height (linear vertical scale). The maximum monthly surge values from 1980 to 2019 were analyzed and the data extrapolated using both linear and polynomial regression models to estimate the likelihood value of storm surge over the next century. While the polynomial regression model distributes weights across multiple variables and their interactions, allowing it to capture more complex relationships within the data, the linear regression model is influenced by the number of data points, especially if the data is not well-distributed. The projections are derived from 882 empirically recorded tide gauges distributed along the global coastlines. The 100-year surge estimate for Rodrigues based on historical data is around 0.34 ± 0.01 m for the linear model and 0.93 ± 0.03 m for the polynomial model. The 100-year surge estimate derived from the linear regression model utilizes the entire dataset and is notably influenced by the pronounced asymmetry towards lower return period values. Conversely, the polynomial fit does not assume a relationship between the lower and higher end values. Both models will be examined in a complementary manner, with the linear regression acting as the baseline model. The results will then be compared to those from the polynomial fit, which captures the non-linear dynamics at higher return period values.

SSPs which represent future climate scenarios were developed based on the assumptions of how population, education, energy use, technology, amongst others, may change over the next century and their related impacts on the global warming temperature and rates of warming (Tebaldi et al. 2021). These were derived using Integrated Assessment Models (IAMs) and are part of the Coupled Model Intercomparison Project Phase 6 (CMIP6). SSPs explore five potential ways the world could develop without climate policies and how vari-

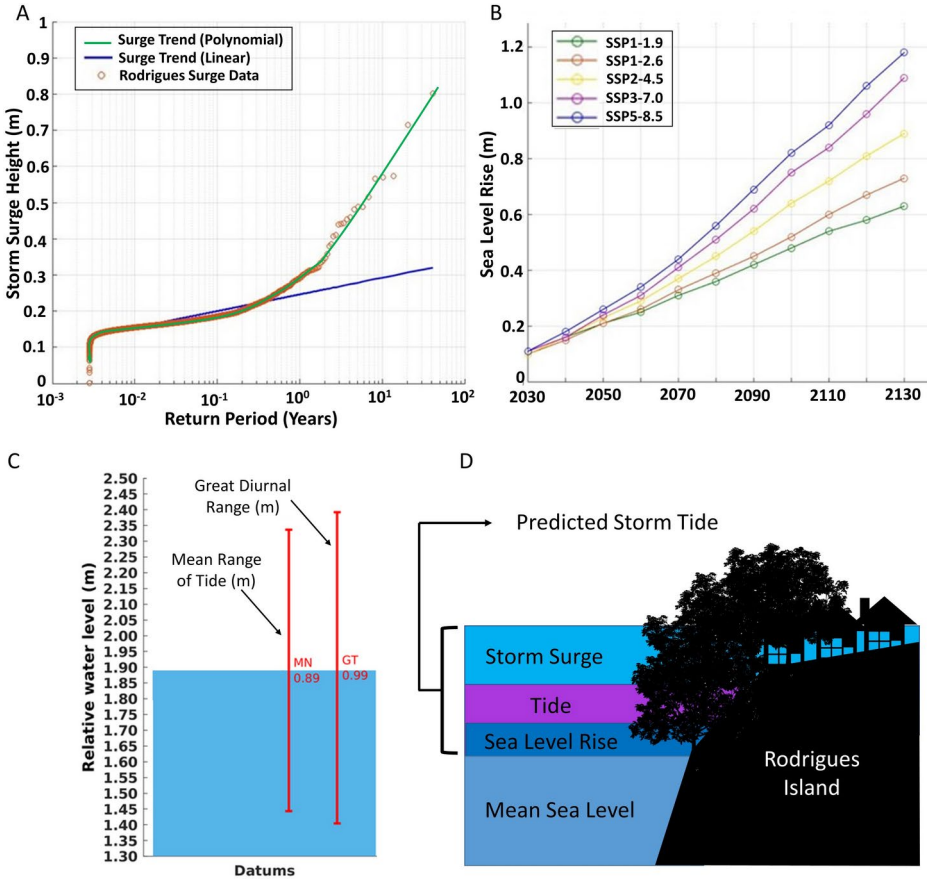

Fig. 2 **a** Return period estimates of storm surge levels on the coast of Rodrigues Island with linear and polynomial regression fits; **b** Projected sea-level rise under five Shared Socio-economic Pathways over the next century; **c** Variations in mean range of tide and great diurnal range about the relative water level in Rodrigues (UHSLC 2024); **d** Schematic diagram illustrating the compounding effect of sea-level rise, astronomical tide, and 100-year storm surge over the next century on the coast of Rodrigues (Adapted from NOAA 2024)

ous levels of climate change mitigation can be achieved by combining the mitigation targets of Representative Concentration Pathways (RCPs) with the SSPs (Meinshausen et al. 2020). The sea-level rise projections under five SSPs scenarios were integrated in the storm tide model. A description of each SSP scenario is presented in Table 1.

The five projected future scenarios from 2030 to 2130 are shown in Fig. 2b, and are taken from the NASA Sea Level Projection Tool (Fox-Kemper et al. 2021). The model implemented by IPCC (Fox-Kemper et al. 2021) provides the most updated regional sea level projections from 2020 to 2150 relative to the baseline period of 1995–2014 (Fox-Kemper et al. 2021). The rising sea levels in the vicinity of Rodrigues Island (19.7245° S, 63.4272° E) were extracted, as presented in Table 2. The projected sea-level rise in Rodrigues for the year 2130 range from 0.63 m under the low-emissions scenario to 1.18 m under the

Table 1 Warming limits of five SSPs investigated in the implementation of the storm surge model (O'Neill et al. 2016)

SSP	Description
SSP1 - 1.9	holds warming to approximately 1.5 °C above 1850–1900 in 2100 after slight overshoot (median) and implies net zero CO2 emissions around the middle of the century
SSP1 - 2.6	stays below 2.0 °C warming relative to 1850–1900 (median) with implied net zero emissions in the second half of the century
SSP2 - 4.5	is approximately in line with the upper end of aggregate Nationally Determined Contribution emission levels by 2030. SR1.5 assessed temperature projections for NDCs to be between 2.7 and 3.4 °C by 2100, corresponding to the upper half of projected warming under SSP2 - 4.5. New or updated NDCs by the end of 2020 did not considerable change the emissions projections up to 2030, although more countries adopted 2050 net zero targets in line with SSP1 - 1.9 or SSP1 - 2.6. The SSP2 - 4.5 scenario deviates mildly from a 'no-additional- climate-policy' reference scenario, resulting in a best-estimate warming around 2.7 °C by the end of the twenty-first century relative to 1850–1900
SSP3 - 7.0	is a medium to high reference scenario resulting from no additional climate policy under the SSP3 socioeconomic development narrative. SSP3 - 7.0 has particularly high non-CO2 emissions, including high aerosols emissions
SSP5 - 8.5	is a high reference scenario with no additional climate policy. Emission levels as high as SSP5 - 8.5 are not obtained by Integrated Assessment Models (IAMs) under any of the SSPs other than the fossil fueled SSP5 socioeconomic development pathway

Table 2 Projected median value of sea-level rise (in metres) for five SPPs from 2030 to 2130

	SSP1 - 1.9	SSP1 - 2.6	SSP2 - 4.5	SSP3 - 7.0	SSP5 - 8.5
2030	0.11 ± 0.04	0.1 ± 0.04	0.1 ± 0.04	0.11 ± 0.04	0.11 ± 0.04
2040	0.16 ± 0.06	0.15 ± 0.06	0.16 ± 0.06	0.16 ± 0.06	0.18 ± 0.06
2050	0.21 ± 0.08	0.21 ± 0.08	0.23 ± 0.08	0.24 ± 0.08	0.26 ± 0.08
2060	0.25 ± 0.10	0.26 ± 0.10	0.29 ± 0.10	0.31 ± 0.10	0.34 ± 0.10
2070	0.31 ± 0.12	0.33 ± 0.12	0.37 ± 0.12	0.41 ± 0.12	0.44 ± 0.12
2080	0.36 ± 0.16	0.39 ± 0.16	0.45 ± 0.16	0.51 ± 0.16	0.56 ± 0.16
2090	0.42 ± 0.20	0.45 ± 0.20	0.54 ± 0.20	0.62 ± 0.20	0.69 ± 0.20
2100	0.48 ± 0.24	0.52 ± 0.24	0.64 ± 0.24	0.75 ± 0.24	0.82 ± 0.24
2110	0.54 ± 0.29	0.6 ± 0.29	0.72 ± 0.29	0.84 ± 0.29	0.92 ± 0.29
2120	0.58 ± 0.34	0.67 ± 0.34	0.81 ± 0.34	0.96 ± 0.34	1.06 ± 0.34
2130	0.63 ± 0.42	0.73 ± 0.42	0.89 ± 0.42	1.09 ± 0.42	1.18 ± 0.42

high-emissions scenario. Projected sea-level rise in 100-year time in the year 2130 varies between 1.18 ± 0.42 m under SSP5 - 8.5 scenario to 0.63 ± 0.42 m under SSP1 - 1.9 scenario.

The Great Diurnal Range (GT) corresponds to the difference in height between the mean higher high water (MHHW) and the mean lower low water (MLLW). It is used to describe the variation in water level over a tidal cycle. This calculation is generally based on continuous water level observations over a specified period, typically 19 years, known as the National Tidal Datum Epoch (NTDE). The University of Hawaii Sea Level Center (UHSLC) employs similar methods to compute tidal datums. This involves collecting con-

tinuous water level data at specific tide stations, processing the data to identify high and low tides, calculating the Mean Higher High Water (MHHW) and Mean Lower Low Water (MLLW) as the average of the higher high-water heights and the lower low water heights, respectively, over the NTDE, and determining the Great Diurnal Range by subtracting the MLLW from the MHHW (NOAA 2003). The great diurnal tide in Rodrigues is estimated to be 0.99 ± 0.05 m based on UHSLC (2024) estimations and as illustrated in Fig. 2c.

The projected storm tide level near Rodrigues in 2130 is calculated by combining the 100-year storm surge level with astronomical tides and rising sea levels, as described by Doorga et al. (2024), and formulated below:

$$Hazard_{stormtide} = S_{surge} + SLR + T \tag{2}$$

where $Hazard_{stormtide}$ is the 100-year storm tide hazard, S_{surge} is the 100-year return period storm surge, SLR is the sea-level rise in 100 years under different SSPs, and T is the distance of the mean higher-high water mark from the mean tide level.

Figure 2d illustrates the compounding effect of various coastal flood factors. Due to the range of scenarios for each factor, including two storm surge regression models (linear and polynomial-based), fiveSSPs (SSP1 - 1.9, SSP1 - 2.6, SSP2 - 4.5, SSP3 - 7.0, and SSP5 - 8.5), and a considerable diurnal tidal range, a matrix of potential sea level states over the next century is shown in Table 3. In the current study, we adopt a 'bathtub approach' to flooding, assuming that the water level will rise uniformly across the coastal landscape, with the coast being hydrologically connected to the sea. It combines the slow onset event of sea-level rise and rapid onset event of storm surge to give an indication of the vertical displacement of water over the next 100 years assuming simultaneous occurrence of events. This method assumes an infinite volume available for flooding, potentially leading to an overestimation of vulnerable areas. Consequently, our results are presented as"low-lying areas exposed to flooding"instead of"inundation extent"to accurately reflect the limitations of our assumptions. Additionally, waves on top of the storm tide could considerably increase the vertical water column, potentially leading to more inland penetration of storm surges and increased coastal flooding. However, this study focuses solely on quantifying the impact of storm tides and does not consider the effects of wind-generated waves. This methodological approach has also been adopted by Doorga et al. (2024) and Sadien et al. (2024).

The potential storm tide levels in 2130 will depend on the warming trajectory and the surge model adopted. In the high-impact scenario, considerable potential damage to coastal infrastructure over the next century is expected to occur under SSP5 - 8.5, where temperatures reach 4.4 °C by the century's end. The high-impact scenario is expected when storm surge return periods are modeled with a polynomial regression model, as it effectively captures the non-linear dynamics at higher return period values. Conversely, the lower-impact scenario is likely under the more optimistic SSP1 - 1.9, with global warming reaching around 1.4 °C by the end of the century. Both scenario limits will be explored in the

Table 3 Matrix showing different sea level states (in meters) in 100 years on the coast of Rodrigues based on the combinations of 100-year storm surge, peak diurnal tide, and rising sea level under five SSPs from the mean tide level

	SSP1 - 1.9	SSP1 - 2.6	SSP2 - 4.5	SSP3 - 7.0	SSP5 - 8.5
Baseline scenario – Linear model	1.46 ± 0.42	1.56 ± 0.42	1.72 ± 0.42	1.92 ± 0.42	2.01 ± 0.42
High surge value scenario – Polynomial model	2.05 ± 0.42	2.15 ± 0.42	2.31 ± 0.42	2.51 ± 0.42	2.60 ± 0.42

current study. The low-impact scenario is anticipated when storm surge return periods are modeled using a linear fit, as this approach assumes data uniformity and is influenced by the entire dataset. Muis et al. (2016) provide a global reanalysis of storm surges and extreme sea levels using the Global Tide and Surge Reanalysis (GTSR) dataset, which is based on hydrodynamic modelling and involves dynamically derived datasets that account for tides and surges. The 100-year return period storm surge estimate for Rodrigues from this dataset is 0.99 m, which aligns closely with the polynomial fit result which is estimated at 0.95 m. This comparison highlights the robustness of the current approach and ensures that the analysis is consistent with established extreme value analysis methods. Although the linear model's 100-return period estimated at 0.34 m is farther from Muis et al.'s (2016) prediction, it remains important for understanding the broader range of potential outcomes. The linear model, while influenced by lower to moderate values and not capturing extreme scenarios, offers a different perspective on storm surge behavior. It helps identify trends and potential impacts under less severe conditions, thereby complementing the insights gained from the polynomial model.

3.2 Vertical datum correction for tide

The average tidal range in Rodrigues is approximately 0.89 m, while the maximum diurnal range is about 0.99 m, as shown in Fig. 2c. Due to the existing tidal variations, it is crucial to adjust the Digital Elevation Model (DEM) of 30 m resolution acquired from Copernicus for Rodrigues to the mean tidal level to ensure accurate storm tide results (European Space Agency 2024). The preprocessing of the DEM involved interpolating elevation values to enhance the coarse 30-m grid resolution dataset. Subsequently, the refined DEM was adjusted to the mean tidal level.

Figure 3 illustrates the tidal correction process. A rectangular tile north of Rodrigues (Fig. 3a) is used as an example. The DEM aligns with the island's coastal profile at the high tide mark (Fig. 3b). The shorelines of Rodrigues at mean and high tide levels were digitized using historical Landsat/Copernicus imagery available on Google Earth. The tidal status (high, low, or mean) was determined based on historical tidal records from Rodrigues, as measured by the Mauritius Meteorological Services (MMS). The historical image of Rodrigues taken on May 9, 2020, at 10:00 am (Fig. 3c) was captured at mean tide level, as indicated by the MMS (2020). In contrast, the image from December 20, 2022, at 10:30 am (Fig. 3d) was taken at high tide (MMS 2022). The shorelines at mean and high tide were overlaid on the DEM, and the missing elevation data between these shorelines were filled

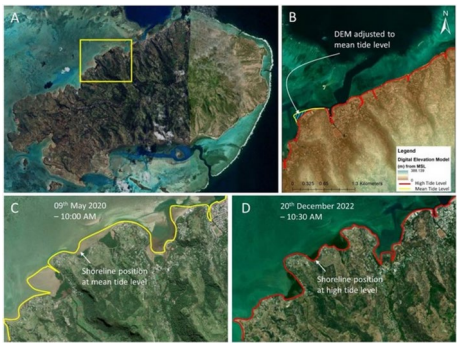

Fig. 3 **a** Map of Rodrigues showing the location of the rectangular tile used as illustrative case for performing vertical datum correction.; **b** Adjustment of the Digital Elevation Model (DEM) relative to the mean tide level in Rodrigues; **c** Shoreline position at mean tide level in the rectangular tile; **d** Shoreline position at high tide level in the rectangular tile

using downscaled measurements from the Shuttle Radar Topography Mission (SRTM) and ASTER Global Digital Elevation Model (GDEM) available on Google Earth. These missing values were interpolated, integrated into the existing DEM, and the scale was readjusted from the baseline of the mean tidal mark.

3.3 Storm tide risk assessment

Storm tide risk ($Risk_{stormtide}$) refers to the likelihood of potential loss or damage occurring due to storm tide hazard ($Hazard_{stormtide}$) impacting coastal elements that are exposed and vulnerable. It may be expressed as:

$$Risk_{stormtide} = Hazard_{stormtide} \times Exposure \times Vulnerability \tag{3}$$

The area affected by storm tides along the coast of Rodrigues which determines the storm tide risk, will be calculated using Eq. (3). Exposure of infrastructure comprising of buildings and roads, to storm tide risk will be assessed through signature analysis and supervised classification of NICFI satellite data at a 4.77 m resolution (Xu 2007). Infrastructure includes critical facilities and systems such as roads and buildings, which are essential for societal functioning. Roads facilitate the movement of people and goods, while buildings provide residential, commercial, industrial, and public services. While water supply systems, electricity grids, and telecommunication systems are crucial components of infrastructure, they will not be analyzed in the current study to maintain a focused scope. However, these elements are recognized for their considerable impact and will be addressed in detail in future research. By concentrating on the key buildings and road infrastructural components, the current study aims to provide a more focused and manageable analysis. Vulnerability, which measures the susceptibility of elements to damage from a given hazard, is assumed to be uniform or negligible in this study due to standardized infrastructure and homogeneous populations. Therefore, vulnerability is not considered. This study aims to estimate storm tide risk based on remotely sensed data of hazard and exposure.

3.4 Data and software

The specifications and characteristics of the data employed in the current study are summarized in Table 4. Geospatial data analysis is conducted on QGIS software (v3.16.8) and Google earth ver.6.0.1.2032 (beta) while computational analysis is done on MS Excel and MATLAB (R2023a).

4 Results

The spectrum of possible flooding extents in 2130 based on the combination of storm surge, sea level rise, and tidal effects is presented in Table 5. The analysis of storm tide impacts using both linear and polynomial surge models reveal considerable differences in projected flooding and infrastructure damage across various SSPs. The linear model indicates a progressive increase in affected areas, with infrastructure damage ranging from 67,384 m^2 to 193,384 m^2 and total flooding from 865,123 m^2 to 1,133,456 m^2. In contrast, the polyno-

Table 4 Specifications of data employed in the current study for coastal storm risk assessment

Data	Source	Time coverage
Storm surge height	Tadesse and Wahl (2021)	1980—2019
Sea-level rise	IPCC (Fox-Kemper et al. 2021)	2030—2130
Astronomical tide	UHSLC (2024)	1986—2017
Satellite image	Google earth ver.6.0.1.2032 (beta) Digital Earth Africa; NICFI	2024
Exposed assets (Buildings and Road)	Google earth ver.6.0.1.2032 (beta) Digital Earth Africa; NICFI	2024
Digital Elevation Model	European Space Agency (2024)	2020

Table 5 Coastal flooding impacts for a 100-year storm tide on the coast of Rodrigues under different SSP scenarios for linear and polynomial surge models

	SSP1 - 1.9	SSP1 - 2.6	SSP2 - 4.5	SSP3 - 7.0	SSP5 - 8.5
Surge Model	**Linear**				
Scenario	Low-impact				High-impact
Building (m^2)	34,389	38,889	42,389	43,305	44,222
Road (m^2)	32,995	83,005	132,995	145,625	158,255
Infrastructure (m^2)	67,384	121,384	175,384	184,384	193,384
Flooding(m^2)	865,123	980,123	1,095,123	1,114,289	1,133,456
Infrastructure/total flooding	7.9%	12.3%	16.0%	16.5%	17.0%
Flooding/total land area	0.80%	0.91%	1.01%	1.03%	1.05%
Surge Model	**Polynomial**				
Building (m^2)	45,139	46,055	46,972	47,889	56,912
Road (m^2)	170,886	183,516	196,146	208,777	284,560
Infrastructure (m^2)	202,384	211,384	220,384	229,384	341,472
Flooding(m^2)	1,152,623	1,171,789	1,190,956	1,210,123	1,446,461
Infrastructure/total flooding	17.5%	18.0%	18.5%	19.0%	23.6%
Flooding/total land area	1.07%	1.08%	1.10%	1.12%	1.34%

mial model predicts a more substantial impact, with infrastructure damage escalating from 202,384 m^2 to 341,472 m^2 and total flooding from 1,152,623 m^2 to 1,446,461 m^2. Notably, the proportion of infrastructure damage relative to total flooding is higher in the polynomial model, peaking at 23.6% under SSP5 - 8.5, compared to 17.0% in the linear model. Additionally, the polynomial model forecasts a greater percentage of total land area affected by flooding, reaching 1.34% under SSP5 - 8.5, as opposed to 1.05% in the linear model.

The extent of storm tide flooding varies, with a lower-level estimate resulting from the combination of a 100-year return period storm surge modeled linearly, sea level rise under SSP1 - 1.9, and the great diurnal tide, impacting coastal amenities and buildings. This combination of sea level conditions will have a relatively lower impact on coastal elements. Conversely, the maximum storm tide flooding occurs when a 100-year return period storm surge modeled using polynomial regression is combined with sea level rise under SSP5 - 8.5, and the great diurnal tide. This scenario results in maximum potential damage to the coastal region of the island over the same 100-year period.

The extent of flooding and impacted infrastructures under both low and high-impact scenarios, assuming inaction by the residents to adapt, are illustrated in Fig. 4. The Rodriguan population is concentrated on the lower-elevated coast, as shown in Fig. 4. Under the high-impact scenario of 2.60 m vertical water displacement (Fig. 4a), the total extent of flooding approximates 1.45 km^2 which is equivalent to 1.34% of total land territory being threatened

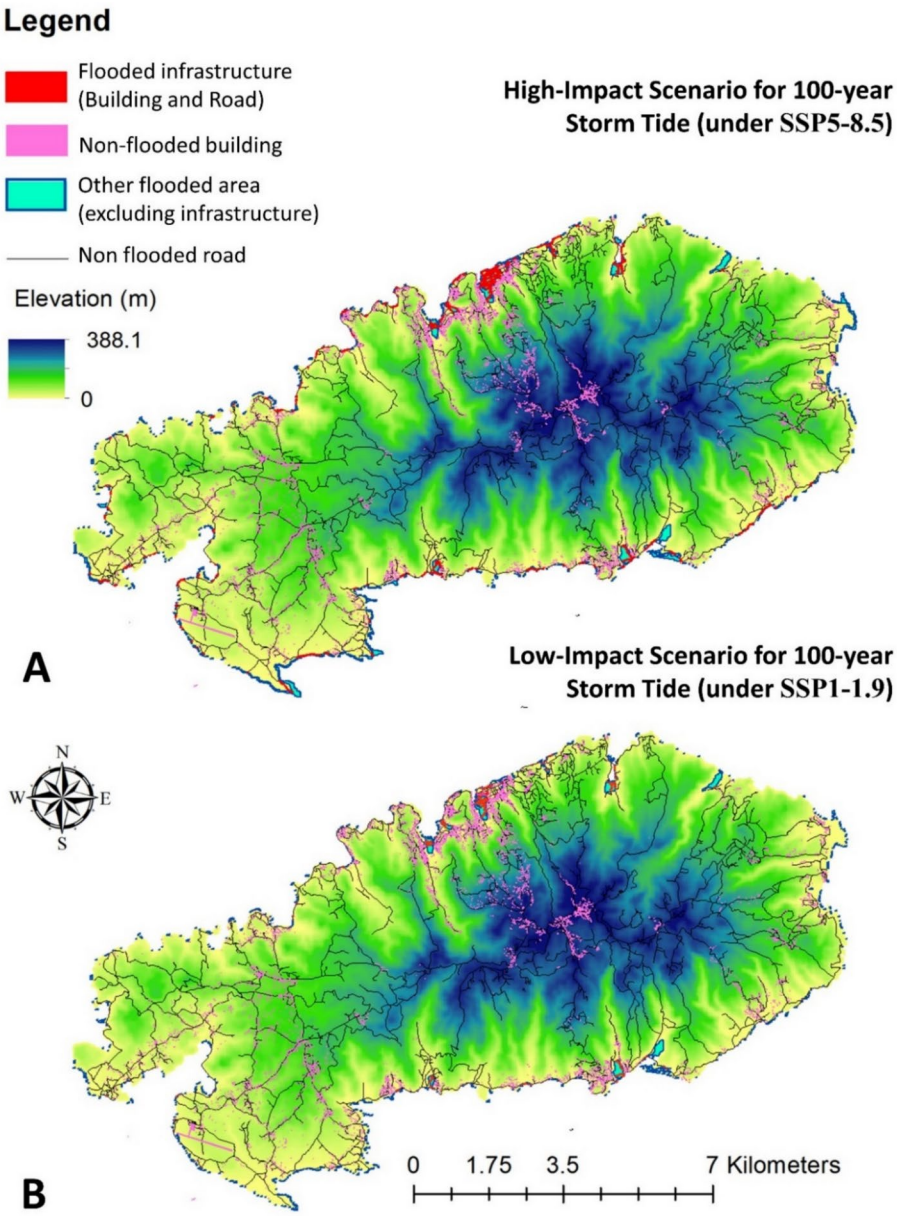

Fig. 4 Coastal inundated areas for a 100-year storm tide under (**a**) the high-impact scenario of SSP5 - 8.5; and (**b**) low-impact scenario of SSP1 - 1.9 in Rodrigues, modelled using a polynomial approach

by the rising and surging ocean. This would result in an infrastructural damage extent of approximately 341,472 m^2, including 56,912 m^2 of building damage and 284,560 m^2 of road damage. Under the low-impact scenario of 1.46 m vertical water displacement (Fig. 4b), the total extent of flooding approximates 865,123 m^2, equivalent to 0.80% of terrestrial land extent lost to sea. This would lead to infrastructural damage extending 67,384 m^2, including 34,389 m^2 of building damage and 32,995 m^2 of road damage.

As illustrated in Fig. 5a, the areas affected by flooding in the low-impact scenario show similar levels of potential damage to both buildings and roads. Under the low-impact scenario, a total of 67,384 m^2 of infrastructural damage is observed, with buildings and roads being equally impacted. However, the scenario changes considerably in the high-impact case. In this scenario, the number of impacted roads exceeds that of buildings due to the higher prevalence of roads in potential flood zones. This results in a notable increase in infrastructural damage, totaling 341,472 m^2. Specifically, road damage is more pronounced because roads typically cover a larger area and are more extensive in flood-prone regions than buildings. This differential impact underscores the need for tailored adaptation strategies. Given the higher vulnerability of roads, it is crucial to enhance the resilience of coastal roads to recurrent storm tide impacts. Measures such as elevating roadways, improving drainage systems, and using flood-resistant materials can considerably reduce the damage and ensure the functionality of critical transportation networks during extreme weather events.

Despite only a slight increase in the ratio of inundated area to total land extent—from 0.80% in the low-impact scenario to 1.34% in the high-impact scenario—the ratio of infrastructure damage to inundated area rises sharply from 7.9% to 23.6%, as shown in Fig. 5b. This disproportionate increase in infrastructural damage relative to the extent of flooding highlights the severe impact of higher water levels. It emphasizes the need for proactive adaptation strategies to minimize damage and safeguard vital infrastructure against future storm tides and sea level rise.

From Fig. 4, it is observed that infrastructural damage is primarily concentrated around the capital area, Port Mathurin. This region, with large portions of land just a few meters above sea level (Fig. 6d), is particularly vulnerable to storm tide flooding and sea level rise. The proximity to the coastline, coupled with the low elevation, makes Port Mathurin highly susceptible to flooding during extreme weather events. In the high-impact scenario, with a vertical water displacement of 2.60 m, considerable portions of Port Mathurin are projected to be flooded. This would lead to extensive damage to buildings, roads, and other critical infrastructure. Although the damage is relatively moderate under the low-impact scenario, it reaches critical levels under the high-impact scenario.

Other regions of considerable infrastructural damage include Baie Aux Huitres, Rivière Cocos, and Anse Mourouk (Fig. 6a, b and e), where infrastructure assets are located in close proximity to the sea, making them vulnerable to coastal flooding. These areas face considerable risk from coastal hazards, with both the built environment and natural ecosystems potentially affected. In the high-impact scenario, substantial portions of these regions are expected to be flooded, leading to extensive damage to roads, bridges, residential buildings, commercial establishments, and tourism infrastructure. The coastal road infrastructure is damaged under the high-impact scenario (Fig. 4a) but is less impacted under the low-impact scenario (Fig. 4b).

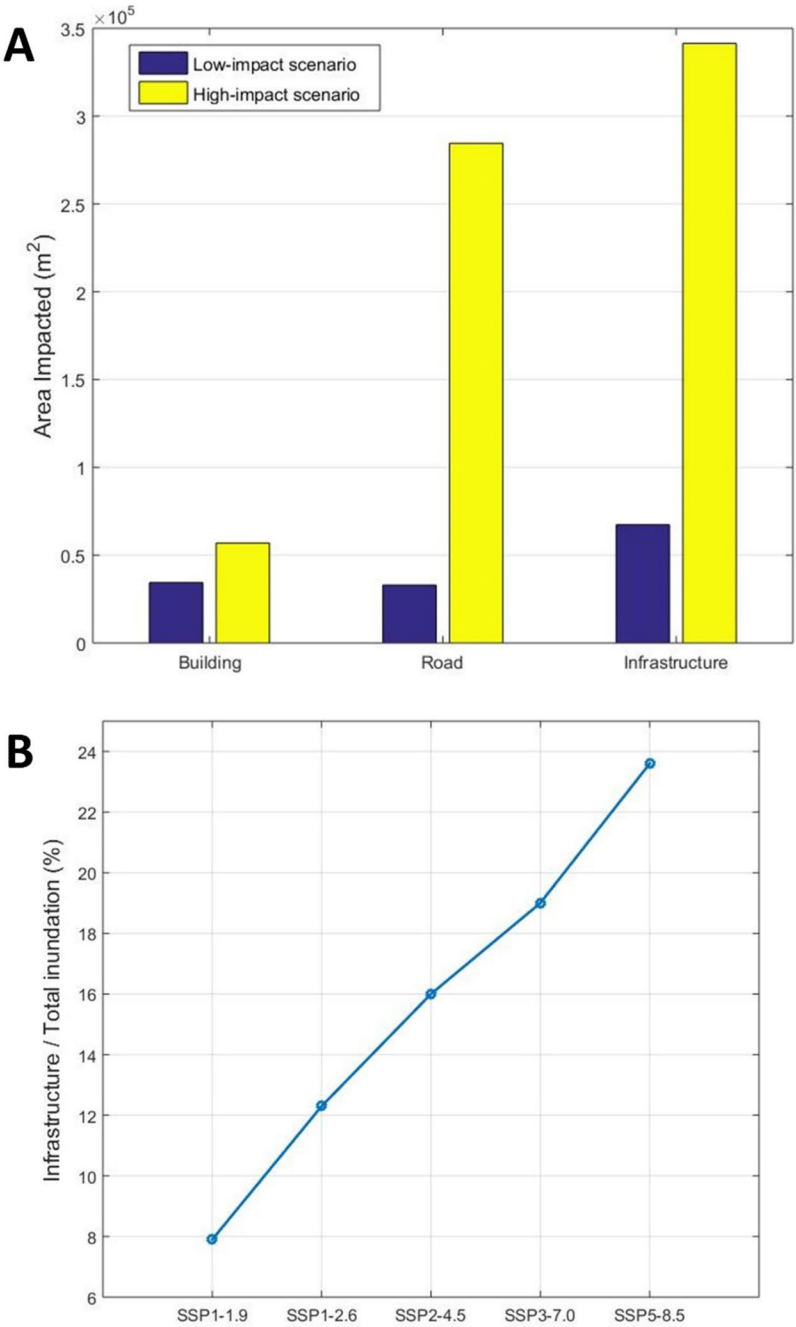

Fig. 5 **a** Barchart illustrating the extent of building, road, and infrastructure impacted under the low-impact linear scenario (SSP1 - 1.9) and high-impact polynomial scenario (SSP5 - 8.5); **b** Trend of infrastructure impacted per total extent of flooding under the linear (SSP1 - 1.9, SSP1 - 2.6, SSP2 - 4.5) and polynomial (SSP3 - 7.0, SSP5 - 8.5) scenarios

Fig. 6 Low-lying coastal regions at risk: **a** Rivière Cocos; **b** Baie Aux Huitres; **c** Grand Baie; **d** Port Mathurin; **e** Anse Mourouk (Source: Authors' illustration)

Recurrent storm surges are expected to impact sand dunes and contribute to coastal erosion, causing the dunes'front face to become steep and scarped. This process reduces the dunes' overall volume and elevation, making them more susceptible to future storms and potential overwash. In Grand Baie, similar patterns of erosion and escarpment are observed (Fig. 6c). The coastal beach profiles in this area, including popular spots like Trou d'Argent and Pointe Coton, are particularly vulnerable under the high-impact scenario. These regions, known for their scenic beauty and tourism facilities, face considerable risks from storm tide hazards. The escarpment formation due to erosion not only affects the natural landscape but also poses threats to infrastructure, such as the hotel at Pointe Coton.

5 Discussions

5.1 Coastal vulnerability

Rodrigues Island is increasingly challenged by climate-induced coastal floodings, worsened by its geographical location and low elevation above sea level. The rising frequency and intensity of storm tides, driven by tropical cyclones, have underscored the island's vulnerability to both immediate and long-term impacts on coastal infrastructure (Douce 2014; Ragoonaden et al. 2017). Recent studies stress the urgent need for comprehensive risk assessments and the development of robust mitigation strategies to protect these coastal regions (Sadien et al. 2024; Doorga et al. 2024). These assessments are essential for understanding the potential extent of damage and for formulating effective adaptation measures.

Figure 6 illustrates low-lying coastal regions vulnerable to storm tide hazards in both high and low-impact scenarios. These five regions share a common feature in that they are located at the base of gentle to steep coastal topography gradients, which heightens the risk of flooding. This increased risk is due not only to storm tide hazard but also to runoff from the mountains in the event of heavy rainfall, which can exacerbate flooding. Studies by Nguyen et al. (2020) for Vietnam and Moraru et al. (2021) for Norway show that

steep slopes contribute considerably to increased runoff, which can exacerbate flooding, especially during heavy rainfall events. Both studies highlight the role of steep slopes in rapidly channeling water into rivers, leading to rapid river rises and severe flooding. This risk is particularly concerning in Rodrigues, where the relatively denser settlements at Port Mathurin and Baie Aux Huitres, and sparsely distributed ones at Rivière Cocos and Anse Mourouk in coastal lowlands (Turner and Klaus 2005), are especially vulnerable to storm tide hazards. Furthermore, Port Mathurin serves as the major area where all major business operations take place, including harbor infrastructure, regional assembly, market area, small businesses, and oil and petrol stockage, among others. Research has shown that storm surge events can have persistent and adverse long-term impacts on the economies of coastal regions. For instance, a study developed a state-level computable general equilibrium model to assess the vulnerability of state and national economies to surge events affecting coastal communities and strategic industrial assets, including petroleum and chemical manufacturing in Texas (Zhang and Lam 2015). The results indicated considerable welfare losses and declines in Gross State Product equivalent to 0.05% due to surge-induced impacts. Furthermore, while the economic impact of storm surges on large coastal states like Texas can be considerable, the disproportionate reliance of small island nations like Rodrigues on their port infrastructure can magnify the economic consequences of such events.

Besides disruption to the port area, the island of Rodrigues is highly dependent on its local economy, which is primarily driven by micro, small, and medium enterprises (MSMEs), including sectors such as agriculture, fisheries, and tourism (Agathe 2024; GoM and ITC 2021). These sectors are crucial for the livelihood of the local population and contribute considerably to the island's GDP. However, the local economy is vulnerable to environmental changes, particularly those induced by climate change. Some of the identified high-risk areas to flooding (Section 4) in Rodrigues are susceptible to coastal erosion (GoM and UNDP 2018). Anthropogenic-induced beach erosion in Rodrigues is however very limited owing to the low level of development of the coastal zone (Duvat 2009). Therefore, most of the damage inflicted to the backshore and dune originate primarily from climate-induced factors, of which rising sea level is a key threat. Closely interlinked with the issue of coastal flooding is the accelerated impacts of coastal erosion. The island of Rodrigues is suffering from severe coastal erosion at localised places around the island. Hard solutions, such as the construction of seawalls, have been implemented to protect coastal communities on the island while limiting coastal erosion (Schoonees et al. 2019). Although these hard engineering solutions can offer immediate protection, they frequently come with long-term disadvantages that may surpass their benefits. These solutions are often criticized for their unattractive appearance and potential to limit beach access for both people and wildlife (Fig. 7). Additionally, they are susceptible to deterioration over time due to continuous exposure to recurring storm surges, and the island’s limited financial resources make it challenging to implement and maintain such high-cost interventions (Klöck et al. 2022). Research has shown that hard structures often lead to maladaptation, redistributing and sometimes exacerbating shoreline erosion rather than stopping it (Barnett and O'Neill 2013).

5.2 Adaptation strategies

The combined effects of a 100-year storm surge, rising sea levels, and tides are anticipated to cause potential damage to coastal infrastructure and land properties, including beach

Fig. 7 Concrete wall along the beach of Anse aux Anglais in Rodrigues to protect the coast against strong swells (Source: Authors' illustration)

areas, as highlighted in the results section. In lieu of the rise in sea level, surging seas and the accompanying threats they may represent, various actions have been taken in Rodrigues Island. Those actions may be classified as nature-based solutions, establishment of Marine Protected Areas (MPAs) and hard structures to limit the impacts of any sea-level rise on the coastal zone. While the effectiveness of hard structures is debatable, as mentioned in Section 5.1, numerous small island states have leaned towards a blend of nature-based solutions, terrestrial-marine conservation, and coastal legal reforms (Nonaka et al. 2024; Suedel et al. 2023; Rohe et al. 2019).

In Rodrigues, ecosystem restoration dates back in the 1980's where a programme was put in place for the planting of mangroves at specific locations within the lagoon with the aim to limit the negative effects of sedimentation occurring from terrestrial runoff (Perry and Berkeley 2009). This programme has proved to be successful with fine sediment being trapped within the mangrove forest instead of being washed into the lagoon (Perry and Berkeley 2009). Furthermore, the mangrove forests established provide numerous ecosystem services to the coastal communities including a natural barrier against the rise in sea level. Mangrove planting and restoration programmes along some of the vulnerable low-lying coastal regions at risk identified in Fig. 6 may be envisaged for enhanced resilience to storm tide hazard. By leveraging natural benefits, a denser network of mangroves can be a sustainable and cost-effective solution for coastal protection through wave energy dissipation, sediment trapping, and storm surge protection (Gracia et al. 2018).

Another nature-based solution being undertaken in Rodrigues Island is coral restoration. It was reported that Rodrigues Island is facing severe bleaching during the period

2023 onwards (Jogee et al. 2023). To limit the impact of climate change on the coral reefs, a coral restoration programme has been initiated on the island. Among the various benefits offered from coral reef ecosystems, it is expected that the newly restored corals will provide a natural barrier against high swells thus limiting the impact of coastal erosion caused by sea-level rise (Roelvink et al. 2021). Rodrigues has five MPAs which covers around 35% of the lagoon waters (UNEP-Nairobi Convention and WIOMSA 2021; Pasnin et al. 2016). Furthermore, two nature reserves have been declared on land covering a total area of 60.5 hectares and another 23 hectares covered under two islets as important bird areas (MWF 2022). Those conservation areas provide protection to the sensitive habitats present and thus enhance the ecosystem services provided by those habitats.

While adaptation measures such as nature-based solutions and Marine Protected Areas are pivotal in addressing the impacts of sea-level rise, the examination of legal frameworks becomes essential to support and institutionalize these measures. To address the unique climate challenges faced by Rodrigues Island, a comprehensive Climate Resilience Ordinance should be established, drawing on frameworks such as the Tonga Climate Change Act 2021. This ordinance would set clear objectives for enhancing climate resilience against sea-level rise, storm surges, and coastal erosion. It would define the roles and responsibilities of local communities, experts from sister Island Mauritius, and NGOs. Essential components would include mandating regular climate risk assessments and detailed risk mapping for vulnerable coastal areas, developing localized adaptation plans based on these assessments, and ensuring active community involvement in the decision-making process to reflect Rodrigues'specific needs. In Rodrigues, where the population predominantly comprises fishermen and farmers, targeted training programs can be implemented to facilitate the cultivation of seagrass and mangroves for coastal protection. To incentivize these efforts, financial rewards could be integrated into the Finance Act 2024 as the form of tax benefits and duty-free part of Payments for Ecosystem Services (PES), subsidies on selected farming products, and green loans similar to Barbados National Biodiversity Action Plan.

The portfolio of solutions must therefore comprise of a blend of nature-based solutions, adequate terrestrial-marine conservation, and coastal resilience strategies coupled with legal provisions. However, in the event that the adaptation strategies are not sufficient, then the solution of managed retreat and relocation of coastal communities may need to be contemplated. As stated by Gussmann and Hinkel (2020), it is key for the government of SIDS to take action to relocate communities at risk from flooding in low-lying vulnerable areas. Yet, past experiences reveal that relocations were primarily undertaken to promote economic development rather than to address rising sea levels or coastal risks (Gussmann and Hinkel 2020). Relocating coastal communities is often politically challenging because these communities are typically unwilling to move, and decision-makers who advocate for such measures face considerable political risks (Gussmann and Hinkel 2020). Proactive relocation of coastal communities must therefore be supported by an implementation strategy package consisting of strong community involvement early in the process, clear communication, financial incentives, infrastructural, legal, and policy framework that support relocation efforts, alongside long-term support to relocated communities. This can be achieved by establishing a legal mandate such as the Fiji's Climate Relocation of Communities Trust Fund Act 2019.

5.3 Limitations

The geospatial information utilized in this study could benefit from enhanced spatial resolution, as higher resolution data would likely yield more precise results. Even after downscaling and refining the digital elevation model, uncertainties remain due to the original dataset's coarseness. Additional uncertainties exist in the constitutive elements of the storm tide model and comprising of uncertainties in the estimation of storm surge, sea-level rise, and astronomical tide. Tadesse et al. (2020) analysis reveals that the model, while effective in certain regions, struggles in tropical areas like Rodrigues. This is due to the limited variability of predictor data in these regions, which hampers the model's ability to accurately capture storm surge patterns. Additionally, the model may overlook crucial processes like wind-wave and swell activity, which are particularly relevant in tropical environments. These limitations underscore the need for caution when interpreting the model's results for Rodrigues, as they may not fully represent the complex dynamics of storm surges in this region. Additionally, the bathtub method does not take into account factors such as land slope, surface roughness, or hydrological connectivity, which can influence the spread and recession of water.

6 Conclusions

In conclusion, the findings in this paper calls for adaptation measures to address the challenges such as rising sea levels, recurrent storm surges and coastal flooding faced by Rodrigues, a small remote SIDS, in the Indian Ocean. The geospatial-based approach employed in this study has provided valuable insights into the specific coastal areas most at risk, serving as a crucial foundation for tailored adaptation strategies. The research used information from the IPCC sea-level projections and combined them with storm surge model information. We find that approximately 1.45 km^2 of terrestrial land would experience flooding under SSP5 - 8.5, representing a loss equivalent to above 1.34% of the island's total land territory at risk from risingsea levels, and inducing infrastructural damage reaching 23.6%. The ratio of infrastructure damage to flooding extent under SSP1 - 1.9, SSP1 - 2.6, SSP2 - 4.5, and SSP3 - 7.0 are estimated at 7.9%, 12.3%, 16.0%, and 19.0%, respectively. In light of the accelerated sea-level rise observed in the forthcoming years, innovative solutions are needed. The last limb of the paper guides the formulation of effective, comprehensive, strategic and place based-adaptation strategies such as nature-based solutions in the form of mangrove planting, coral restoration, and terrestrial conservation, establishment of Marine Protected Areas and engineering solutions. A review of the legal framework is also mentioned, with a focus on relocation, and to support Nature-based solutions. This research, in sum, presents a contribution to the understanding and adaptation of climate change impacts in Rodrigues Island to safeguard its rich natural heritage, whilst requesting concerted efforts, collaboration and strategic interventions among policymakers.

Acknowledgements The authors acknowledge the support provided by the Université des Mascareignes in the provision of facilities for conducting the research presented in this paper. The authors extend their gratitude to Shoals Rodrigues for providing the photographs of the shoreline of Rodrigues. The use of QGIS, Google Earth, and MATLAB for research purposes is also acknowledged. The authors also wish to thank the anonymous reviewers and editors for their valuable feedback and contributions to improving the quality of

this research. The scientific results and conclusions, as well as any views or opinions expressed herein, are those of the authors and do not necessarily reflect the views of the individual data contributors.

Author contributions Conceptualization: JD.
Methodology: JD.
Investigation: JD, OP, YD.
Visualization: JD.
Writing: JD, OP, YD.

Data availability The data used in the current research are open-source.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Agathe H (2024) Implication of tourism development in Rodrigues: ecotourism and sport tourism in a sustainable perspective. In: sport tourism, island territories and sustainable development: a comparative perspective. Cham: Springer Nature Switzerland, pp 393–412

Apollonio C, Bruno MF, Iemmolo G, Molfetta MG, Pellicani R (2020) Flood risk evaluation in ungauged coastal areas: The case study of Ippocampo (Southern Italy). Water 12(5):1466

Barnett J, O'Neill SJ (2013) Minimising the risk of maladaptation: a framework for analysis. Climate adaptation futures, pp 87–93

Chapman B, Turner JR (2004) Development of a geographical information system for the marine resources of Rodrigues. J Nat Hist 38(23–24):2937–2957

Doorga JR, Sadien M, Bheeroo NA, Pasnin O, Gooroochurn O, Modoosoodun-Nicolas K, Ramchandur V, Ramharai D (2021) Assessment and management of coastal erosion: insights from two tropical sandy shores in Mauritius Island. Ocean Coast Manag 212:105823

Doorga JRS, Bernardie-Tahir N, Deenapanray PN, Dindoyal Y, Mycoo M, Moncada S (2024) Surging seas, rising sea levels, and sinking communities: The urgent need for climate adaptation in small island states. Environ Sci Policy 157:103788

Douce Y (2014) Coastline impacts of tropical cyclones and climate change on Mauritius (Doctoral dissertation, University of KwaZulu-Natal, Durban)

Duvat V (2009) Beach erosion management in small island developing states: Indian Ocean case studies. Coastal Process 126:149–160

European Space Agency (2024) Copernicus global digital elevation model. Distributed by OpenTopography. https://doi.org/10.5069/G9028PQB. Accessed 2024-12-04

Farbotko C (2010) Wishful sinking: disappearing islands, climate refugees and cosmopolitan experimentation. Asia Pac Viewp 51(1):47–60

Fox-Kemper B, Hewitt HT, Xiao C, Aðalgeirsdóttir G, Drijfhout SS, Edwards TL, Golledge NR, Hemer M, Kopp RE, Krinner G, Mix A, Notz D, Nowicki S, Nurhati IS, Ruiz L, Sallée J-B, Slangen ABA, Yu Y (2021) Ocean, cryosphere and sea level change. In: Masson-Delmotte V, Zhai P, Pirani A, Connors SL, Péan C, Berger S, Caud N, Chen Y, Goldfarb L, Gomis MI, Huang M, Leitzell K, Lonnoy E, Matthews JBR, Maycock TK, Waterfield T, Yelekçi O, Yu R, Zhou B (eds) Climate change: the physical science basis. Contribution of working group i to the sixth assessment report of the intergovernmental panel on climate change. Cambridge University Press

Glas H, Jonckheere M, Mandal A, James-Williamson S, De Maeyer P, Deruyter G (2017) A GIS-based tool for flood damage assessment and delineation of a methodology for future risk assessment: case study for Annotto Bay, Jamaica. Nat Hazards 88:1867–1891

GoM and ITC (2021) Mauritius National Export Strategy Rodrigues Island Strategy 2017–2021

GoM and UNDP (2018) Vulnerability & Adaptation Assessment (VAA) Toolkit (Mauritius): User Manual for Coastal Zone Sector

Gracia AD, Rangel-Buitrago N, Oakley JA, Williams AT (2018) Use of ecosystems in coastal erosion management. Ocean Coast Manag 156:277–289

Gussmann G, Hinkel J (2020) What drives relocation policies in the Maldives? Clim Change 163(2):931–951
Hardman ER, Blais FEI, Desiré MS, Raffin JSJ, Perrine S, Raffaut R, Chinien-Chetty M (2006) Annual report of benthos, reef fish and invertebrate surveys for reef slope and reef flat areas in Rodrigues 2005. Shoals Rodrigues unpublished report, p 44
IPCC (2007) Climate change 2007: the physical science basis. In: Solomon S, Qin D, Manning M, Chen Z, Marquis M, Averyt KB, Tignor M, Miller HL (eds) Contribution of working group I to the fourth assessment report of the intergovernmental panel on climate change. Cambridge University Press, Cambridge, United Kingdom and New York, p 996
Jogee SY, Gopalsing S, Jeetun S, Ricot M, Taleb-Hossenkhan N, Mattan-Moorgawa S, Kaullysing D, Wijayanti DP, Casareto BE, Suzuki Y, Bhagooli R (2023) First report of diseases and compromised health conditions on hard corals around Rodrigues Island, Southwest Indian Ocean. Diversity 15(10):1086
Klöck C, Nunn PD (2019) Adaptation to climate change in small island developing states: a systematic literature review of academic research. J Environ Dev 28(2):196–218
Klöck C, Duvat VK, Nunn PD (2022) Maladaptive diffusion? The spread of hard protection to adapt to coastal erosion and flooding along island coasts in the Pacific and Indian Ocean. Reg Environ Change 22(4):136
Knutson T, Camargo SJ, Chan JC, Emanuel K, Ho CH, Kossin J, Mohapatra M, Satoh M, Sugi M, Walsh K, Wu L (2020) Tropical cyclones and climate change assessment. Bull Am Meteor Soc 101(3):E303–E322
Lee H, Romero J (2023) Climate change 2023: synthesis report. In: A report of the intergovernmental panel on climate change. Contribution of working groups I, II and III to the sixth assessment report of the intergovernmental panel on climate change. IPCC, Geneva, Switzerland, p 24
Lincke D, Hinkel J (2021) Coastal migration due to 21st century sea-level rise. Earth's Future 9(5):e2020EF001965
Magnan AK, Oppenheimer M, Garschagen M, Buchanan MK, Duvat VK, Forbes DL, Ford JD, Lambert E, Petzold J, Renaud FG, Sebesvari Z (2022) Sea level rise risks and societal adaptation benefits in low-lying coastal areas. Scientific Reports 12(1):10677
Mauritius NDC (2021) Available online at https://faolex.fao.org/docs/pdf/mat216390.pdf. Accessed on 24/2/24. Accessed 20 Feb 2024
McGranahan G, Balk D, Anderson B (2007) The rising tide: assessing the risks of climate change and human settlements in low elevation coastal zones. Environ Urban 19(1):17–37
McMichael C, Katonivualiku M (2020) Thick temporalities of planned relocation in Fiji. Geoforum 108:286–294
Meinshausen M, Nicholls ZR, Lewis J, Gidden MJ, Vogel E, Freund M, Beyerle U, Gessner C, Nauels A, Bauer N, Canadell JG (2020) The shared socio-economic pathway (SSP) greenhouse gas concentrations and their extensions to 2500. Geosci Model Dev 13(8):3571–3605
MMS (2020) mauritius meteorological services tidal predictions for Rodrigues - May 2020. http://metservice.intnet.mu/mmsimages/May%20-June%20%202020%20TideRod.pdf
MMS (2022) Mauritius meteorological services tidal predictions for Rodrigues - December 2022. http://metservice.intnet.mu/mmsimages/01%20December%202022-%20January%202023%20TideRod.pdf
Moraru A, Pavlíček M, Bruland O, Rüther N (2021) The story of a steep river: causes and effects of the flash flood on 24 july 2017 in western Norway. Water 13(12):1688
Muis S, Verlaan M, Winsemius HC, Aerts JC, Ward PJ (2016) A global reanalysis of storm surges and extreme sea levels. Nat Commun 7(1):11969
MWF (2022) Annual report on the activities of the Mauritian Wildlife Foundation. Available online: https://www.mauritian-wildlife.org/mwf-files/files/files/Mauritian%20Wildlife%20Foundation%20Annual%20Report%202022.pdf. Accessed 20 Feb 2024
Mycoo MA (2018) Beyond 1.5 C: vulnerabilities and adaptation strategies for Caribbean Small Island developing states. Reg Environ Change 18(8):2341–2353
Needham HF, Keim BD, Sathiaraj D (2015) A review of tropical cyclone-generated storm surges: global data sources, observations, and impacts. Rev Geophys 53(2):545–591
Nguyen BD, Minh DT, Ahmad A, Nguyen QL (2020) The role of relative slope length in flood hazard mapping using AHP and GIS (case study: Lam River Basin, Vietnam). Geogr, Environ, Sustain 13(2):115–123
NOAA (2003) Computational techniques for Tidal Datums Handbook. NOAA Special Publication NOS CO-OPS 2. https://tidesandcurrents.noaa.gov/publications/Computational_Techniques_for_Tidal_Datums_handbook.pdf
NOAA (2024) What is storm surge? https://oceanservice.noaa.gov/facts/stormsurge-stormtide.html
Nonaka A, Tajima Y, Kono T, Yamano H, Orita T, Kayanne H (2024) Nature-based Solutions (NbS) for sustainable coastal communities in the Pacific Island Countries and Territories. In: Sustainable development across Pacific Islands: lessons, challenges, and ways forward. Springer Nature Singapore, Singapore, pp 227–253

O'Neill BC, Tebaldi C, Van Vuuren DP, Eyring V, Friedlingstein P, Hurtt G, Knutti R, Kriegler E, Lamarque JF, Lowe J, Meehl GA (2016) The scenario model intercomparison project (ScenarioMIP) for CMIP6. Geosci Model Dev 9(9):3461–3482

Pasnin O, Attwood C, Klaus R (2016) Marine systematic conservation planning for Rodrigues Island, western Indian Ocean. Ocean Coast Manag 130:213–220

Pielke Jr. RA, Gratz J, Landsea CW, Collins D, Saunders MA, Musulin R (2008) Normalized hurricane damage in the United States: 1900–2005. Natural Hazards Review 9(1):29–42

Perry CT, Berkeley A (2009) Intertidal substrate modification as a result of mangrove planting: Impacts of introduced mangrove species on sediment microfacies characteristics. Estuar Coast Shelf Sci 81(2):225–237

Petzold J, Magnan AK (2019) Climate change: thinking small islands beyond Small Island Developing States (SIDS). Clim Change 152(1):145–165

Ragoonaden S, Seewoobaduth J, Cheenacunnan I (2017) Recent acceleration of sea level rise in Mauritius and Rodrigues. West Indian Ocean J Mar Sci 51–65

Robinson SA (2020) Climate change adaptation in SIDS: a systematic review of the literature pre and post the IPCC Fifth Assessment Report. Wiley Interdiscip Rev: Clim Change 11(4):e653

Roelvink FE, Storlazzi CD, Van Dongeren AR, Pearson SG (2021) Coral reef restorations can be optimized to reduce coastal flooding hazards. Front Mar Sci 8:653945

Rohe JR, Govan H, Schlüter A, Ferse SC (2019) A legal pluralism perspective on coastal fisheries governance in two Pacific Island countries. Mar Policy 100:90–97

Sadien M, Doorga JR, Rughooputh SD (2024) Assessment of tangible coastal inundation damage related to critical infrastructure and buildings: the case of Mauritius Island. Int J Dis Risk Reduction 114:104909

Saxena A, Qui K, Robinson SA (2018) Knowledge, attitudes and practices of climate adaptation actors towards resilience and transformation in a 1.5 C world. Environ Sci Policy 80:152–159

Scandurra G, Romano AA, Ronghi M, Carfora A (2018) On the vulnerability of Small Island Developing States: a dynamic analysis. Ecol Ind 84:382–392

Schoonees T, Gijón Mancheño A, Scheres B, Bouma TJ, Silva R, Schlurmann T, Schüttrumpf H (2019) Hard structures for coastal protection, towards greener designs. Estuaries Coasts 42:1709–1729

Suedel BC, Metcalfe CD, Emmanuel N, Frederick N (2023) Nature-based solutions for building resilience in small island developing States in the Caribbean. In: Building Resilience to Climate Change in Small Island Developing States (SIDS) in the Caribbean. Cham: Springer Nature Switzerland, pp 35–48

Tadesse MG, Wahl T (2021) A database of global storm surge reconstructions. Scientific Data 8(1):125

Tadesse M, Wahl T, Cid A (2020) Data-driven modeling of global storm surges. Front Mar Sci 7:260

Tebaldi C, Debeire K, Eyring V, Fischer E, Fyfe J, Friedlingstein P, Knutti R, Lowe J, O'Neill B, Sanderson B, Van Vuuren D (2021) Climate model projections from the scenario model intercomparison project (ScenarioMIP) of CMIP6. Earth Syst Dyn 12(1):253–293

Thomas A, Baptiste A, Martyr-Koller R, Pringle P, Rhiney K (2020) Climate change and small island developing states. Annu Rev Environ Resour 45:1–27

Turner J, Jago C, Daby D, Klaus R (2000) The mascarene region. In: Sheppard CRC (ed) Seas at the millenium: environmental evaluation, Elsevier Science Ltd. pp 243–258

Turner J, Klaus R (2005) Coral reefs of the Mascarenes, western Indian Ocean. Philos Trans R Soc A: Math, Phys Eng Sci 363(1826):229–250

UHSLC (2024) University of Hawaii Sea Level Centre– Water Level. Available online at https://uhslc.soest.hawaii.edu/stations/?stn=105#levels. Accessed 20 Feb 2024

UNDP (2021) Coastal life and resilience diversifying livelihoods and building resilience of coastal communities on Rodrigues Island, Mauritius. https://undp.shorthandstories.com/gef-sgp-coastal-life-and-resilience/index.html

UNEP-Nairobi Convention and WIOMSA (2021) Western Indian Ocean marine protected areas outlook: towards achievement of the global biodiversity framework targets. UNEP and WIOMSA, Nairobi, Kenya, p 298

Walshe RA, Stancioff CE (2018) Small island perspectives on climate change. Island Stud J 13(1):13–24

Wang S, Mu L, Yao Z, Gao J, Zhao E, Wang L (2021) Assessing and zoning of typhoon storm surge risk with a geographic information system (GIS) technique: a case study of the coastal area of Huizhou. Nat Hazard 21(1):439–462

Wong PP (2011) Small island developing states. Wiley Interdisc Rev: Clim Change 2(1):1–6

Xu H (2007) Extraction of urban built-up land features from Landsat imagery using a thematicoriented index combination technique. Photogramm Eng Remote Sens 73(12):1381–1391

Zhang Y, Lam JSL (2015) Estimating the economic losses of port disruption due to extreme wind events. Ocean Coast Manag 116:300–310

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Authors and Affiliations

Jay Rovisham Singh Doorga[1] · Olivier Pasnin[2] · Yeshna Dindoyal[3]

✉ Jay Rovisham Singh Doorga
jdoorga@udm.ac.mu

[1] Faculty of Sustainable Development and Engineering, Université Des Mascareignes, Rose Hill, Mauritius

[2] Reef Conservation, Les Flammants Branch Road Morcellement Pereyscape, Péreybère, Mauritius

[3] Stratton Court, Level 4, Room 26, Poudrière Street, Port Louis, Mauritius